NED University of Engineering and Technology
Department Of Software Engineering
CEP Description Form

Course Code and Title: SE-312 Software Construction and Development

Select any one: Problem Based Learning✓/Open Ended Lab/Design Project

PROGRAMME LEARNING OUTCOMES COVERED	
PLO-2	Problem Analysis

COGNITIVE LEVELS COVERED	
C2	Comprehension

PROBLEM STATEMENT:

Design and develop a software system that addresses a real-world challenge related to sustainable development. The system should provide solutions to optimize the use, management, or distribution of critical resources such as water, energy, food, or transportation in a way that promotes sustainability, efficiency, and long-term impact. The problem should be framed based on one or more of the UN Sustainable Development Goals (SDGs).

REQUIREMENTS:

- Identify and define a specific problem related to sustainability that the software will solve.
- Perform a detailed problem analysis, identifying stakeholders, key requirements, and constraints.
- Propose a software solution, including high-level design, architecture, and technologies involved.
- Develop and implement the system following all phases of software construction, including design, coding, testing, and deployment.

DELIVERABLES:

1. **Problem Definition Document (PDD)**:
    - Description of the specific problem related to sustainability.
    - Stakeholder analysis.
    - Functional and non-functional requirements.
    - Key assumptions, constraints, and risks.

2. **System Architecture and Design**
    - High level architectural diagrams.
    - Detailed design document including class diagrams, database schema, and user interfaces.

3. **Project Plan:**

- Work breakdown structure (WBS).
- Timeline and milestones for each phase of development.
- Risk management plan.

4. Software Implementation:

- Fully functional codebase of the developed software.
- Demonstration of the working system.

5. Testing Plan and Test Cases:

- Unit, integration, and system testing with detailed test cases.
- Bug report and how issues were resolved.

6. Deployment Strategy Document:

- Deployment environment and platform.
- Deployment process (manual/automated), including any dependencies.
- Backup, recovery, and security measures.

7. Maintenance and Scalability Plan:

- Monitoring and logging strategies.
- Bug tracking and issue resolution workflows.
- Update process for new features, patches, or improvements.
- Scalability strategy to handle increased load or additional features.

DEPARTMENT OF SOFTWARE ENGINEERING
BACHELORS IN SOFTWARE ENGINEERING
Course Code: SE-204
Course Title: Software Construction and Development
<u>Complex Engineering Activity</u>
SE Batch 2024, Spring Semester 2026
Grading Rubric

Group Members:

Student No.	**Name**	**Roll No.**
S1		
S2		
S3		

CRITERIA AND SCALES			**Marks Obtained**		
			S1	**S2**	**S3**
Criterion 1: Does the problem statement/ motivation clearly stated?					
0 – 1	2 - 3	4 – 5			
Objective of the project are not clear or described.	Some lack of clarity in objectives/purpose.	The project's objectives are clearly stated.			
Criterion 2: How well is the design and architecture structured and presented?					
0 – 1	2 - 3	4 – 5			
Incomplete or inconsistent design; architecture does not address core needs and is poorly structured.	Basic design that meets most requirements; architecture is effective with minor improvement areas.	Exceptional design with innovative solution; architecture is highly efficient, scalable, and well-documented.			
Criterion 3: How well is the code organized and maintainable?					
0 – 1	2 - 3	4 – 5			
Code is poorly organized, lacks modularity, and has man errors; difficult to maintain.	Code is functional but lacks clarity, modularity or structure; several practices are not followed.	Code is exceptionally well written, clean, modular and full adheres to best practices; highly maintainable.			
Criterion 4: How effectively is error handling and system resilience implemented?					
0 – 1	2 - 3	4 – 5			
Minimal error handling; system is vulnerable to failures or crashes.	Good error handling; covers most scenarios but lacks coverage of extreme cases.	Comprehensive error handling with robust recover mechanism; system remain stable under all conditions.			
Criterion 5: How thorough is the student's testing and validation?					
0 – 1	2 - 3	4 – 5			

Minimal testing with limited coverage; insufficient documentations of results and errors.	Basic testing with some major cases covered but inadequate validation of edge cases; documentation lacking.	Exhaustive testing with thorough documentation; all critical cases and edge scenarios are rigorously validated.			
Criterion 6: How effectively is the deployment strategy and maintenance plan outlined?					
0 – 1	2 - 3	4 – 5			
Deployment strategy is unclear and manual; maintenance plan lacks coherence and detail.	Good deployment strategy but lacks some automation; maintenance plan exists but is not detailed.	Comprehensive, automated deployment strategy with clear, detailed maintenance plan ensuring long term success.			
Criterion 7: How did the student participate individually in a team?					
0 – 1	2 - 3	4 – 5			
The individual did not contribute to the project and failed to meet responsibilities. The individual does not identify key performance criteria of successful teams or draw inference to own experience	The individual did not contribute as heavily as others but did meet all responsibilities. The individual is also able to identify some key performance criteria of successful teams and/or draw related connections the group performance	The individual contributed in a valuable way to the project. The individual is also able to articulate the key performance criteria of successful teams and evaluate the group performance accordingly			
		Total Marks:			

Teacher's Signature:_____________